2.4

Tuesday, October 8, 2024 9:46 AM

Permutation Matrix

↳ Have a 1 in every row and a 1 in every column. All other entries are zero.

↳ Also known as row-swapping matrix.

For example → $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ $\left.\begin{matrix} a \to c \\ b \to a \\ c \to b \end{matrix}\right\}$ This way I want to do swappings

So, $Px = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} c \\ a \\ b \end{bmatrix}$

↳ Basically the rows you want to swap, swap those rows in identity matrix.

PA = LU Factorization

An example:

$$A = \begin{bmatrix} 1 & 2 & a \\ 2 & 4 & b \\ 3 & 7 & c \end{bmatrix} \xrightarrow{E} \begin{bmatrix} 1 & 2 & a \\ 0 & 0 & b-2a \\ 0 & 1 & c-3a \end{bmatrix} \xrightarrow{P} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & c-3a \\ 0 & 0 & b-2a \end{bmatrix} = U$$

↓

this violate the property of lower triangular matrix. In order to make it L, row operation has to be applied.

↳ $PA = \begin{bmatrix} 1 & 2 & a \\ 3 & 7 & c \\ 2 & 4 & b \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}}_{L}\underbrace{\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & c-3a \\ 0 & 0 & b-2a \end{bmatrix}}_{U}$

Transpose Matrix

↳ Swapping row and column

if $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ then $A^T = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 4 \end{bmatrix}$

Then, $(A^T)_{ij} = A_{ji}$

↳ $(AB)^T = B^T A^T$. Also $(ABC)^T = C^T B^T A^T$.

↳ $(A^{-1})^T = (A^T)^{-1}$. Also $((A)^T)^T = A$.

↳ $(A+B)^T = A^T + B^T$ and $(A-B)^T = A^T - B^T$

Symmetric Matrices

if $A = A^T$ → A is a symmetric matrix.

For example: $S = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = S^T$.

Example: $A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$

Which permutation makes PA upper triangular?

Which permutation makes $P_1 A P_2$ lower triangular?

$$PA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \rightarrow U$$

$$P_1 A P_2 = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{P_1} \underbrace{\begin{bmatrix} 6 & 0 & 0 \\ 5 & 4 & 0 \\ 1 & 2 & 3 \end{bmatrix}}_{A \rightarrow L} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{P_2}$$

$\Rightarrow$ When you want to swap rows multiply from left side. PA will swap the rows of A.

When you want to swap columns multiply from right side. AP will swap the columns of A.

Example: Find the $PA = LU$ Factorization.

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

Sol^n:

$$PA = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\Rightarrow PA = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow E_{31} PA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow E_{31} PA = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & -3/2 & -1 \end{bmatrix} \qquad R_3 = R_3 - \tfrac{1}{2} R_1$$

$$\Rightarrow E_{32} E_{31} PA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3/2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & -3/2 & -1 \end{bmatrix}$$

$$\hookrightarrow E_{32}E_{31}PA = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1/2 \end{bmatrix} \quad R_3 = R_3 + \frac{3}{2} R_2$$

$$\Rightarrow PA = E_{31}^{-1} E_{32}^{-1} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$\Rightarrow PA = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1/2 \end{bmatrix}}_{U}$$